

EDUCATION	Advanced College of Engineering and Management Institute of Engineering (IOE), Tribhuvan University <i>B.E. in Computer Engineering, Rank: 2 / 96</i> Kathmandu, Nepal 2023
	• Advisor: Prof. Dr. Subarna Shakya • Research area: Recommender System and Machine Learning
PUBLICATIONS	1. Wang Y, Sikka H, Mathur Y*, Zhou T*, Nyachhyon J*, Guruprasad P. GUI-Perturbed: Domain Randomization Reveals Systematic Brittleness in GUI Grounding Models. NeurIPS '26 workshop (under review). (<i>*Equal Contribution</i>) [DOI] 2. Recchia G, Mangat C, Nyachhyon J, Sharma M, Canavan C, Epstein-Gross D, Abdulbari M. Confirmation bias: A challenge for scalable oversight. AAAI-AIA, 2026. [DOI] 3. Nyachhyon J, Sharma M, Thapa P, Bal BK. Consolidating and Developing Benchmarking Datasets for the Nepali Natural Language Understanding Tasks. ACL-IJCNLP, 2025. [DOI] 4. Thapa P*, Nyachhyon J*, Sharma M*, Bal BK. Development of Pre-Trained Transformer-based Models for the Nepali Language. (CHIPSAL) COLING, 2025. (<i>*Joint First Author</i>) [DOI]
TALKS & PRESENTATIONS	• <i>Consolidating and Developing Benchmarking Datasets for the Nepali Natural Language Understanding Tasks, ACL-IJCNLP 2026</i>, Conference Presentation. • <i>Development of Pre-Trained Transformer-based Models for the Nepali Language, COLING 2025</i>, Oral Presentation.
RESEARCH EXPERIENCE	Manifold Research Group <i>Open Science Research Fellow</i> USA (Remote) 2025.10 - 2026.04
	• Software Control Agents: Built and benchmarked LLM-driven software control agents on Mind2Web, conducting systematic comparisons of models (e.g., UI-TARS, Qwen2.5-VL) in complex, real-world web environments. • Trajectory Generation: Developed trajectory generation and perturbation pipelines for training and stress-testing agents, enabling robustness evaluation under distribution shifts and unseen task variants. • Analysis of Agent Behavior: Conducted systematic analysis of agent behavior, focusing on spatial reasoning, tool use, and action grounding failures.
	Information and Language Processing Research Lab (ILPRL) <i>Research Assistant, Supervisor: Dr. Bal Krishna Bal</i> Kavre, Nepal 2024.07 - 2025.12
	• Benchmarking: Led the design and release of the NLUE benchmark (12 Nepali NLU tasks), establishing a standardized evaluation suite for low-resource LLMs. • Nepali Corpus for Foundation Models: Assembled and processed 27.5GB Nepali text corpus addressing data scarcity. • Pre-Training and Analysis: Pretrained BERT/RoBERTa/GPT-2 with instruction tuning, achieving SOTA on Nep-gLUE; conducted comparative evaluation and error analysis.

Institute for Research and Innovation in Intelligent Systems (IRIIS)	Nepal
Co-founder & Researcher	2024.12 - Present

- **Agentic Benchmark Framework:** Designing a formal evaluation framework for LLM agents with an explicit episode schema and multi-dimensional scoring, moving beyond single-number task success rates.
- **Multilingual Planning Evaluation:** Constructing a benchmark for multistep instruction planning over real procedural domains, with gold solutions stored as partial-order DAGs to credit valid alternative orderings.

Modulo Research Ltd.	Cambridge, UK (Remote)
Research Intern, Supervisor: Dr. Gabriel Recchia	2024.06 - 2024.08

- **Automated Analysis:** Automated analysis of 20-minute screen recordings, refining scene change detection with 80% precision via URL change tracking and LLM-based event and detail extraction of video frames.
- **LLM Alignment Evaluation:** Conducted scalable oversight experiments by comparing LLM outputs with human annotations to assess and improve intent alignment.

PROFESSIONAL EXPERIENCE	Insyde AI	Maryland, USA (Remote)
	AI Developer	2025.01 - Present

- **LLM Agent Systems:** Developed production-grade LLM agents for financial workflows with multi-step reasoning, tool use, and automated decision pipelines.
- **MCP & Memory:** Integrated Model Context Protocol (MCP) servers, external tools, and persistent memory for context-aware assistants.
- **Workflow Automation:** Reduced customer processing time by >90%, increasing operational throughput by over 10x.

Virtly IT & Business Solutions Sarl (ICEBRKR)	Geneva, Switzerland (Remote)
ML Engineer	2024.03 - 2024.09

- **Finetuning:** Finetuned BART, T5, Pegasus for summarization (ROUGE-L > 50); Phi-3 for task prioritization system.
- **Scheduling Algorithm:** Built an algorithm to resolve online meeting scheduling conflicts by proposing optimal time slots.

LogicTronix	Lalitpur, Nepal
ML/CV Engineer	2023.06 - 2024.03

- **3D Object Detection & Tracking:** Integrated and optimized SFA3D into ADAS stack, achieving real-time 3D object detection and tracking on embedded platforms.
- **Model Quantization:** Reduced model size by over 60% through quantization for Xilinx FPGA deployment, improving edge inference efficiency.

ACADEMIC EXPERIENCE	Herald College Kathmandu, University of Wolverhampton	Kathmandu, Nepal
	Assistant Lecturer	2026.03 - Present

- **Teaching Delivery:** Deliver tutorials and workshops spanning classical deep learning through to modern agentic AI systems, integrating mathematical foundations with hands-on implementation.
- **Practical Guidance:** Design and supervise implementation-based learning (e.g., MNIST and Devanagari datasets, agent-based automation builds), mentoring group and individual work emphasis on correct methodology and consistency.

VOLUNTEER

PyTorch Tester & Mentor

DeepLearning.AI

Remote

2025.08 - 2026.02

- **Bug Hunting:** Test courses that includes lectures, quizzes, labs and assignments for general and code specific bugs, and report them.
- **Mentorship:** Guide learners on PyTorch and deep learning, resolving quiz conflicts, clarifying doubts, and supporting forum discussions in DeepLearning.AI courses.

Instructor & Event Organizer

Advanced College of Engineering and Management

Kathmandu, Nepal

2022.03 - 2023.01

- **Workshop:** Led hands-on workshops mentoring 100+ peers and juniors in programming and ML fundamentals.
- **Curriculum Development:** Developed a structured curriculum covering Python (basic to advanced) and introductory Machine Learning.
- **Techfest:** Organized international zonal qualification from Nepal for IITB Techfest.

REFERENCES

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Prof. Dr. Subarna Shakya

Director, Information Technology Innovation Center, Tribhuvan University

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